# Supplementary Evidence: When Do LLM Side-Channel Defenses Protect Agent Workflows?

This supplement provides the search accounting, corpus maps, saved empirical results and source-bound audits accompanying the main paper. The package README maps these sections to machine-readable data and the fit-free verifier. Original failed and negative outcomes are preserved alongside positive results.

## S1. Search and Evidence Audit

TABLE S1. ROUTE ACCOUNTING FOR THE BOUNDED CODED SET. COUNTS OVERLAP AND DO NOT FORM A COMPLETE PRISMA FLOW.

| Route / disposition | Count |
|---|---|
| Broad IEEE Q4 hits; not fully screened | 35,743 |
| Amended Q4b export; recall not verified | 847 |
| IEEE saved export occurrences / distinct records | 2,079 / 2,001 |
| IEEE full-text referrals / background routes | 25 / 205 |
| ACM saved occurrences / publication groups | 16,687 / 13,201 |
| ACM Oct. 14 provisional exclusions / full-text referrals / unresolved | 9,368 / 883 / 2,950 |
| ACM dispute groups / combined check candidates (Oct. 18) | 332 / 57 |
| 57-item source check: scoped dispositions / unresolved | 57 / 0 |
| Backward reference occurrences (40 fixed seeds) | 2,237 |
| Forward seeds resolved / distinct returned works | 38 / 24 |
| Recent finite reconciliation: overlapping decision rows | 109 |
| Source-limited candidates not admitted / background chapter | 4 / 1 |
| Final core / adjacent studies | 43 / 17 |
| Final coded study families | 60 |

The 57 scoped dispositions are not completed full readings. B069's complete official abstract supports background routing; later-page eligibility remains unchecked. The 109 rows overlap; 883/2,950 are dated global routes. Q4b does not replace unscreened Q4. Coded-set cutoff: October 8, 2026.

Targeted discovery beyond native exports. The saved September 12 query families include `"agent" "traffic" "fingerprinting"` with an arXiv domain filter, `"MCP" "timing" "side channel"` with the same filter, and `"LLM" "side-channel" "2026"` with USENIX/NDSS filters. October 8 follow-up includes `"agent" "side channel" "2026" privacy workflow` and exact-title checks for AgentPrint, UI Traces and FP-Agent. Primary PMLR records were also checked for prompt-cache evidence. These web-discovery routes yielded source leads, not enumerated native hit exports or a venue-completeness denominator. The companion search logs distinguish discovery, primary-source inspection and subsequent family decisions; no later source lookup is relabeled as native query recall.

Acquisition and grouping. Table S2 prints the actual native queries. Export grouping uses normalized DOI, then normalized title only without a conflicting explicit DOI; study-family/version merging is a later source decision. The acquired IEEE frame has 2,079 occurrences/2,001 distinct records, and ACM has 16,687 occurrences/13,201 groups. ACM final-named exports were byte/hash matched; the Q4b June and Q5 count discrepancies remain unresolved. Historical broad Q4 membership cannot be reconstructed from its later first-1,000 prefix.

Finite accounting versus coverage. Every prepared metadata ID has a disposition: 4,753 unique IDs comprise 2,778 exclusions, 116 referrals and 1,859 unresolved routes. Original answer artifacts include 4,653 schema-valid IDs, 80 format holds and 20 missing finals. Later source reconciliation of 1,775 IDs distinguishes 1,313 containers, 455 scoped dispositions and seven material holds; it is not 455 full readings. The 109-row nominated source map overlaps these and citation routes. The complete row-level decisions and source scopes are in the companion package; none of these frames is a subtractive full-search PRISMA flow.

Known papers and exclusion checks. An exact-identity replay against the saved 2,001-record IEEE and 13,201-group ACM frames finds 1/24 and 0/24 fixed known papers. Online indexing and native field eligibility remain unverified. Literal title/abstract term matching does not emulate native stemming or indexed fields: even the retrieved Early Bird record fails that strict literal test for its saved branches. We therefore report export presence, not recall or evidence that nonhits are ineligible.

The earlier frozen 1,555-record exclusion frame has 1,300 checks (1,075 keyword-risk and 225 stratified): 1,297 retained abstract exclusions, two referrals and one missing abstract. One referral was later excluded against a preprint; two material checks and 255 unsampled lower-risk records remain. Separate later checks cover 33/554, 16/144 and 20/2,090 routes, yielding 3, 2 and 5 source/material holds. The last covers only 10/1,074 keyword-risk records. These AI-assisted checks are not interchangeable frames and do not estimate false negatives. The 23-case blinded AI check

TABLE S2. EXECUTED NATIVE QUERIES AND SAVED EXPORT OCCURRENCES. IEEE DATES: OCTOBER 8, 2026; ACM QUERY/PARTITION ACQUISITION: OCTOBER 8–9, 2026. COUNTS ARE NOT INCLUDED STUDY FAMILIES.

| ID | Exact query | IEEE saved/hits | ACM saved/original hits |
|---|---|---|---|
| Q1 | `("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")` | 63/63 | 479/479 |
| Q2 | `(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")` | 671/671 | 6,181/6,181 |
| Q3 | `("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)` | 37/37 | 511/511 |
| Q4 | `("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)` | 0/35,743 | Not run |
| Q4b | `(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))` | 847/847 | 4,026/4,026 |
| Q5 | `("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")` | 95/95 | 3,689/3,688 |
| Q6 | `("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")` | 366/366 | 1,801/1,801 |

IEEE used Command Search/default metadata, without a lower-year or document-type filter; every result page was exported as RIS with Citation and Abstract. Q4's zero means not exported in the October 8 acquisition, not zero hits or a completed screen. Q4b supplements Q4; recall preservation is unverified. The later Q4 prefix does not reconstruct historical membership. ACM used the All field in the ACM Full-Text Collection (not Guide), then native E-Publication Date partitions and All Results/BibTeX export. Partition receipts retain exact dates and URLs. Q4b's June bin has 285 displayed versus 284 saved; agreement of the overall saved/original counts does not repair the missing identity. Q5's partition total differs from its original whole-query count. Final-named export files were byte/hash matched on October 9; these two retrieval discrepancies remain. The October 8 coded-set cutoff is checked separately from acquisition partitions.

has 17/21 determinate agreement; the 332-dispute readbacks yield 57 source-scoped dispositions, not human double coding.

Final source dispositions and counterconditions. Four source-limited candidates are not admitted to the present core analysis: Wiretapping LLMs (primary full text not obtained); Entropy Based Obfuscation (publisher PDF saved, but adapted/simulated tables lack established new-result lineage); the memory-bandwidth poster (empirical report not obtained); browser-panel analysis (auxiliary versus direct disclosure not established from accessible primary material). B069 is background on its complete official abstract; later pages were not obtained. These are final dispositions for this search, not assertions that unavailable evidence does not exist.

The source checks seek counterconditions to broad deployment claims: AHB's non-Fake online point prevents extrapolating one combined-condition task loss; UI's matched delayed-view adaptation preserves a positive source-local judgment; SpliceDefense and the cache prototypes preserve applicable component evidence. Unacquired broad results and incompletely audited exclusions limit coverage. The query/partition receipts, source dispositions and exact known-paper checks preserve that boundary without treating another search as a prerequisite to reporting the supported comparisons.

## S2. Family Index, Corpus Maps and Prior Work

The accompanying *Supplementary Evidence* archive is the access route for the full tables and fit-free checks. Its README starts with the current paper, family index, result ledger, utility/cost inventory and claim links. Each result joins a citation-key family, source version/hash and locator to its receiver/intervention fields and supported judgment; Table S3 is a filled example. Separate case-evidence entries identify additional source subconditions, rather than silently pooling them into the 33 scoped inventory records.

The index enumerates 60 families: 43 core (15 agent-native, 28 runtime/application-inherited) and 17 adjacent. Four core components are diagnostics; the other 39 do not automatically establish unauthorized confidentiality leakage. Mixed semantic attribution remains adjacent under the auxiliary-only rule. The package preserves all 33 scoped result records/66 field positions and the complete original empirical and workflow analyses. Source locators and hashes support inspection; third-party full texts/raw traces are obtained from their named sources.

Roles of the remaining indexed sources. Additional core studies concern content/hidden-property inference [1], [2], [3], [4], [5], [6], identity/provenance attribution [7], [8], [9], [10], and authorized integrity diagnostics [11], [12], [13]. Adjacent audit, execution, browser-data and profiling studies supply boundary comparisons rather than additional auxiliary privacy attacks [14], [15], [16], [17], [18], [19], [20], [21]. The index records each receiver, target relation and evidence scope; the reference list includes these indexed sources, not just the central case papers.

### S2.1. RQ sensitivity

The expansion adds procedural skills, physical receivers, topology shifts and architecture timing to RQ1–2. RQ3 still

TABLE S3. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

| Join key / field | Entry |
|---|---|
| Family / result | `dangol2026workflow` / `DANGOL-livebinding`. |
| Version / locator | arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [22]. |
| Target / view | Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation. |
| Intervention | Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures. |
| Workload / testbed | Workflow classes with client, relay, gateway and agent co-located on one loopback host. |
| Outcome / cost | Accuracy 0.30; about 13× bytes and 11× latency in that condition. |
| Unresolved fields | Objective terminal task utility NR-R; precise paired live-refit comparison NR-V. |
| Licensed judgment | These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter. |

requires generator-specific bridges. RQ4 separates simulation, component overhead, fixed guardrails and executed protection. Section S6 identifies which specific conclusion disappears when a load-bearing source is removed.

## S3. Retained Failure-Inclusive Controls

The 288-trace fixed-policy pilot has call-count-determined labels and saturated coarse F1=1.00, so it supports no sequence-increment claim. In the separate tool-front-end study, the observer sees both tools; the constrained baseline succeeds on 19/24 tasks, below the frozen promotion gate. These local results are not passive-network or deployed-defense evidence.

The retail case retains 148 windows, 49 empty observations and 102 stopped-error episodes. All three frozen loss contrasts have intervals including zero; test goal achievement is 7/40 versus 8/40 and joint utility remains unresolved. Neither this null-compatible estimate nor failed tasks establish absence of leakage or effective defense. Full protocols and results remain in the delivered evidence package.

## S4. Public-Trace Reanalysis: Protocol and Boundaries

UI Traces (commit `4f3df18`) supplies 4,200 records/14 declared identities, retaining 1,806 error/absent-result indicators. Labels are declarations, not authenticated backends or task correctness. The supplemental package binds source bytes, code and receipts.

Predictors use JS types, local time and callback order, excluding collector time/navigation, URL/DOM text, questions, answers and diagnostics. Cross-document order assumes an archive receiver, not authenticated website capture. Exact-question-disjoint train/validation/test groups have 2,043/1,022/1,135 rows and 395/202/226 questions (823 distinct questions). The source split identifies FRAMES but has no semantic task-family field. Full identity–dataset/exact-question support and a seven-bin first-word *grammatical proxy*, explicitly not a semantic family, are supplemental; neither semantic independence nor family balance is established.

The original 19-fit comparison selects weighted logistic/forest candidates on validation loss only; selected models are forests. H2 adds type bigrams to coarse counts/page-age. Original intervals use 3,000 paired-question resamples and a nominal 98.333% three-contrast allocation, conditional on fixed fitting/selection and noise realizations. Archived H1's identical selected predictor yields exactly zero difference and interval $[0, 0]$; it is a selection illustration, not equivalence evidence.

For $Q$ questions with $n_q$ records each, loss is $Q^{-1}\sum_q n_q^{-1}\sum_{i\in q} -\log p_{i,y_i}$; accuracy substitutes the correct-argmax indicator. Fit weights $N/(Qn_q)$ equalize question weight. Add $\epsilon = 10^{-6}$ to every class probability, renormalize, then clip true-label probability to $[\epsilon, 1]$. Bootstrap draws $Q$ questions with replacement; original contrasts share draw indices, with percentile endpoints 0.8333% and 99.1667%.

Controls declared with original outcomes known. The additive protocol is frozen before new fits, with original outcomes known. All-14 common support is empty. Select the largest identity subset with at least 20 common training questions, then greatest support and lexical tie-break: Gemini 3.1, Gemma 4-31B, Seed 2-lite, with 32/10/10 questions. Hash order selects one row per question/identity, retaining failures; test has 30 rows. This is a three-class sensitivity result.

The first-Wikipedia-document *proxy* stops at HTTP navigation, beforeunload, unmarked origin/URL change or clock reset, but not SPA history alone. Markers gate membership and never enter features. No frame/document IDs authenticate membership; late callbacks may cross boundaries. Empty views remain and epochs are not pooled. The supplement gives the complete six-event projection example and boundary implementation.

Three row-seeded shuffles preserve counts/timestamps/coarse features in *all partitions*, refitting combined candidates with coarse retained. Fits total 8 matched-task, 8 document-proxy, 8 clean and 12 shuffled; all 36 finish before test prediction, without convergence warnings. Clean H2 reproduces exactly. Each shuffle selects coarse: zero increment is then an algebraic consequence of selecting the same predictor. This checks the selection pipeline, not whether shuffled sequence features contain no information; a forced-sequence comparison would be a different, unperformed analysis.

New intervals use 3,000 question resamples, seed 20261018 and 98.333% allocation across the three new contrasts, conditional on fits and shuffle streams. They are exploratory, not blind confirmation or global adjustment across all historical analyses. Separate code reconstructs points/intervals from saved probabilities. Matched-task un-

TABLE S4. PRIMARY OBSERVABLE/RECEIVER PROFILE × SIGNAL ORIGIN IN THE 43 CORE FAMILIES. COLUMN $n$ COUNTS DISTINCT FAMILIES; EACH FAMILY OCCUPIES ONE PROFILE ROW BUT MAY HAVE MULTIPLE ORIGINS. ORIGIN CELLS COUNT FAMILIES, NOT RESULTS, AND THEREFORE NEED NOT SUM TO $n$. EXTRA OBSERVER VIEWS REMAIN IN THE LEDGER. ZEROS ARE NOT FIELD-WIDE GAPS.

| Observable / receiving permission | $n$ | P1 | P2 | P4 | P5 | P6 | P7 | P8 | Representative source |
|---|---|---|---|---|---|---|---|---|---|
| Flow/destination | 12 | 0 | 8 | 0 | 5 | 1 | 2 | 2 | AgentPrint; WRA; NetEcho |
| Own-site actions/attributes | 5 | 0 | 0 | 0 | 5 | 0 | 5 | 0 | UI Traces; FP-Agent |
| API duration/cache/oracle | 12 | 0 | 12 | 0 | 0 | 0 | 0 | 0 | Early Bird; LeakyLMs |
| Resource/physical | 10 | 10 | 3 | 0 | 1 | 0 | 0 | 0 | TDXRay; ClawGuard; LLM RCA (diagnostics) |
| Trajectory/procedure/log | 4 | 0 | 0 | 1 | 4 | 2 | 0 | 0 | SIGLEAK; cycle detection (diagnostic) |
| Total distinct families | 43 | Each family contributes once to $n$. | | | | | | | |

P1 hardware; P2 serving; P4 application/API; P5 orchestration; P6 tools; P7 environment; P8 multi-agent. P3 denotes transport; this collection codes it as a receiving layer but has no P3 generating-origin entry. This does not rule out transport-generated signals elsewhere. Composite origins are multi-label descriptions, not independent channels. Privileged logs and physical diagnostic receivers are not ordinary website capabilities.

TABLE S5. FROM PRIOR CROSS-LAYER SYSTEMATIZATIONS TO RESULT-LEVEL DEPLOYMENT DECISIONS: THE SPECIFIC COMPARISON OBJECT AND ITS DEMONSTRATED USE.

| Prior work | Established perspective | Added result-level question | Demonstrating analysis |
|---|---|---|---|
| Banerjee et al. [23] | Cross-layer compound AI attacks can relax threat assumptions; MITRE organization | Does cover end at the protected interface, get reserialized, or execute as an action? | NetEcho final-hop check and Dangol/AHB transport/action distinction; explicit conditions for permitting transfer. |
| Forough et al. [24] | Confidential-computing threat model; partial inference-defense transfer; isolation is not side-channel immunity | Which oracle, execution mode and component-cost result licenses the proposed defense transfer? | Early Bird KV/semantic split and KVGov hardware/simulation/analytic separation. |
| Cai et al. [25] | Traffic features, shaping objectives and bandwidth trade-offs | Can a traffic-cover cost/utility argument be used for a dummy tool action? | Dangol/AHB: a state-machine boundary determines which task/policy outcomes must additionally be measured. |
| Debenedetti et al. [26] | Auxiliary ML components induce privacy channels | Distinguish an execution-mediated oracle from ordinary semantic attribution | C3 boundary and mixed-output eligibility judgments. |
| Lahjouji–Colaco [27] | Agent data surfaces, governance, composition and cross-session risks | Which bridge test would license a specific transfer? | Receiver/intervention and generator/holdout result maps. |
| Kim; Horváth et al. [28], [29] | Broad agent security; physical DNN mechanisms/threat models | Compare source and receiving privileges without a universal strength order | Remote calibration versus own-site, local and privileged diagnostic profiles. |

certainty and the 509/1,135 test error flags (3.4–91.5% by identity) prevent intrinsic-identity claims.

The saved-result verifier supplies local replay; it does not establish independent replication or live utility/cost.

Failure/activity diagnostics. The configuration was frozen with prior outcomes known: three representations, four candidates each, original weights, seed 20261008; all 12 fits precede test prediction. Failure-only uses error/absent-result bits; empty-only uses no projected events; activity-only uses log-count plus seven page-age summaries, not cross-document duration. Candidate hyperparameters and projections are supplemental. No new confirmatory intervals are supplied. Flags cover 43.00% overall and 44.85% test rows, not authenticated task success. Stratifying 509 flagged/203 questions versus 626 unflagged/216 questions changes support; increments are not a causal failure effect. Semantic-family balance remains unavailable.

Exact H3 perturbation and adaptation. The unchanged perturbation sets $t'_{ire} = t_{ie} + u_{ire}$, $u_{ire} \sim U(0, 5000\,\mathrm{ms})$, independently per valid page-local timestamp in source-array order. Missing times consume no draw; types/order/page keys remain fixed. Three row-seeded realizations transform all partitions. Each adaptation clones the clean-selected representation/architecture and fits once on perturbed training rows with identical weights, without retuning or additional data (three fits). Both programs predict the same perturbed test rows; average paired losses across realizations, rows within question, then questions. The supplemental exact protocol supplies entry points, seed strings and invariants. The transformation edits stored times; it does not execute action delays.

## S5. Intervention Inventory and Evidence Dispositions

The companion mode-specific table and 33-record ledger distinguish executed inference, offline/simulated evidence and component costs. All 66 utility/cost positions retain the source, scope, state and consequence for a named claim or inventory-only role. The main deployment-question table does not treat these heterogeneous records as a common success/failure denominator.

Four textual defense assertions lack the quantitative binding needed for a matched effect (Table S12). Their un-

TABLE S6. SOURCE–RECEIVER–INTERVENTION MAP. THE TARGET COLUMN DISTINGUISHES CONTENT/PROPERTY INFERENCE, ATTRIBUTION AND HUMAN–AGENT DISTINCTION; NONE ALONE ESTABLISHES AN UNAUTHORIZED PRIVACY VIOLATION. U DENOTES AN UNRESOLVED SOURCE/RUN BINDING.

| Source / target | Signal origin → receiver | Intervention / evaluated mode | Surviving path or restriction | Comparison consequence |
|---|---|---|---|---|
| UI Traces / identity attribution | Agent actions → own-site JS | Recorded timestamp delays; offline D2 | Types and order preserved | Test residual action features; no live task-cost estimate. |
| NetEcho / content inference | Serving → network A/B traces | Upstream padding/batching; protected-data fit | Decoded upstream content/state can be reserialized downstream | Inspect final-hop emission; D1 is not automatically D2. |
| SIGLEAK / skill nondisclosure | Procedure → task/output receiver | Common prompt defense | Procedural signatures remain | Compare same receiver/target; not a paired installation effect. |
| AHB / human–agent distinction | Touch behavior → app/detector | Executed fake touches; separate transforms | Task state may change; fit binding U | Measure transitions and objective utility; do not import offline delay costs. |
| WRA / private-intent inference | Real/decoy research requests → visited domains | Concurrent execution; interleaved-domain evaluation | Intent reconstruction falls with one decoy | Preserve domain-view privacy benefit; terminal decoy utility and cost are separately unreported. |
| Dangol / workflow-property inference | Protocol/traffic → defined link | Stored captures versus live OHTTP (§9.7, OHTTP paragraph after Table 5) | Feature ablation differs from network masking | Keep accuracy 0.17 capture and 0.30 live populations separate. |
| Early Bird / cache state | Cache reuse → API timing | KV simulation versus GPT-Cache prototype | Distinct matching mechanisms | 6ms/4% applies to semantic prototype only. |
| KVGov / private prompt | Shared cache → probe oracle | Simulated inference; real timing; analytic isolation | Hash-key isolation not semantic-index isolation | Hardware gap does not become deployed defended ASR. |
| FP-Agent / identity attribution | Policy/browser → visited site | Task/site evaluation, mixed attributes | Static configuration may remain | Task holdout is not independent-visitor holdout. |
| Traffic benchmark / topology | Multi-agent workflow → selected links | Explicit topology/view shift | Unseen generators differ from hidden links | Cross topology with receiving-view restriction. |
| LeakyLMs / architecture | Inference → streaming client | Chosen-prompt timing; no measured defense | Same-GPU-class calibration for architecture | API reachability is not hardware-independent recovery. |

TABLE S7. DEPENDENCY–HOLDOUT–CLAIM MAP. EACH ROW EXCLUDES ONE EXPLANATION, NOT ALL CONFOUNDING.

| Dependency / source | Required holdout | Claim licensed; residual limit |
|---|---|---|
| Task / UI Traces | Question/task-family | New task units; fixed harness remains. |
| Browser / FP-Agent | Visitor/configuration | Configuration transfer; task-only split does not test it. |
| Workflow / traffic benchmark | Agent topology or composition | Tested topology shift; no all-orchestrator guarantee. |
| Hardware / LeakyLMs | GPU/serving calibration | Tested architectural transfer given calibration; not unknown-hardware recovery. |
| Repository / coding fingerprints | Repository and generation pathway | Tested attribution domain; not covert human-mediated use. |
| Order / worked case | Same units, refit shuffle | Fixed-program order increment; not task-independent identity. |

TABLE S8. FIXED-ARCHIVE PREDICTIVE EVIDENCE AND MATCHED SUPPORT. UNITS: NATS OF QUESTION-WEIGHTED LOSS REDUCTION. POSITIVE VALUES FAVOR THE SECOND PROGRAM. INTERVALS CONDITION ON FITTED PROGRAMS; THE ORIGINAL AND ADDITIVE-CONTROL ALLOCATIONS REMAIN SEPARATE.

| Contrast | Estimate | 98.333% interval |
|---|---|---|
| Worked case: coarse vs. combined (H2) | 0.3447 | [0.2974, 0.3944] |
| Matched questions, 3 identities[a] | 0.2634 | [−0.2584, 0.7241] |
| Reanalysis: clean vs. adapted (H3)[b] | 0.1503 | [0.1205, 0.1815] |

[a] 30 rows/10 test questions, 3 classes, rather than 1,135 rows/226 questions, 14 classes; not a direct effect-size comparison. Its interval uses the additive three-control allocation. Archived identical-predictor H1 remains in the protocol.
[b] Both programs predict the same timestamp-perturbed test rows; three fixed realizations are averaged.
Cross-document order assumes an archive receiver, not verified ordinary-website capture. Intervals vary test-question weights, not training, task families or deployments. No live defense utility or cost is inferred.

bound quantities are excluded from comparison; their qualifying attack evidence remains. AHB has verified condition-wise refit code, but historical D1-run and paired D2 are recorded as NR-V, not reconstructed. Its task outcome is not inferred from that code.

D1 denotes protected-condition fitting/calibration; D2 denotes a paired training-procedure comparison on the same defended test condition. Neither entails feasible labeled-data access or matched budgets. Token payload, plaintext padding, wire bytes, model loading, map lookup and added service latency remain distinct cost objects. A scoped NR-

TABLE S9. PUBLIC-TRACE PERMISSION PROJECTION. "SITE" MEANS ITS OWN DOCUMENT ONLY; IT DOES NOT AUTHENTICATE CROSS-DOCUMENT CAPTURE. FULL FIELD-LEVEL CSV AND ONE COMPLETE INGESTED-RECORD PROJECTION ARE SUPPLEMENTAL.

| Field | Site JS | Analysis use |
|---|---|---|
| Type, page-local time | Yes | Original predictors; activity diagnostic |
| Within-document order | Yes | Type bigrams |
| Cross-document order | Unverified | Archive-receiver bigrams |
| URL/origin | Own page | Origin gate only; no text/page-key feature |
| HTTP navigation, collector time | No | Boundary proxy only / time excluded |
| Error/result metadata | No | Analyst-only failure baseline |
| Task/answer metadata | No | Excluded; question hash splits/weights only |
| DOM/link text | Own page | Excluded from all predictors |
| Document/frame ID | Absent | No authenticated document membership |

TABLE S10. POST-REVIEW H2 CONTROLS. POSITIVE VALUES DENOTE A QUESTION-MEAN LOG-LOSS REDUCTION (NATS); BRACKETS ARE CONDITIONAL 98.333% RESAMPLING INTERVALS. NOT A NEW AGENT RUN.

| Contrast | Test rows / questions | Effect [interval] |
|---|---|---|
| Matched tasks, 3 identities: coarse minus sequence | 30/10 | 0.2634 [−0.2584,0.7241] |
| First-document-epoch *proxy*: coarse minus sequence | 1135/226 | 0.1504 [0.1193,0.1825] |
| Original clean minus mean of 3 refit shuffle increments | 1135/226 | 0.3447 [0.2961,0.3953] |

All 14-class common support is empty. The first-document view is reconstructed, not authenticated. Its 0.1504 feature increment and H3's 0.1503 timestamp-adaptation effect are distinct contrasts on the same archive, not independent replications. Cross-document order assumes an archive receiver. All three shuffles select coarse, so their zero increments make this contrast's 0.3447 point equal to original H2. The intervals use separate frozen bootstrap seeds (20261018 here; 20261008 for original H2), yielding different finite-resampling endpoints. Zero selected increments do not establish information equivalence. No task utility or live defense is measured.

R completes the requested extraction, rather than asserting zero cost or preserved utility.

## S6. Conclusion Sensitivity

Table S14 exposes direct-source dependence. Removing NetEcho withdraws the only downstream re-emission instance; removing AHB withdraws the measured mobile state-change instance, while WRA's privacy-oriented decoy result remains. Removing WRA withdraws that private-intent decoy result while AHB retains the task-state branch. UI Traces and SIGLEAK remain different surviving-feature instances, not replications. Removing Dangol withdraws its capture/live contrast; Early Bird, KVGov and SpliceDefense retain separate component examples.

Starting from 60/43/17 total/core/adjacent families, deleting AHB, WildAuth and coding fingerprints leaves 57/42/15; deleting four diagnostics leaves 56/39/17; joint removal leaves 53/38/15. These are current-set ablations. The four latest additions change diagnostic/semantic-boundary coverage but are not premises of the cover/action contrast. For the five multifeature-trace studies, unresolved bindings bound D1 at three–five and D2 at two–five; removing AHB gives three–four and two–four. Early Bird calibration is a separate stage.

Excluding all four unbound defense quantities leaves the three deployment cases unchanged, because none uses those quantities as a premise. AHB's detector-run unknowns prevent an adaptive-detector claim but do not erase its separately reported task-success points. Its non-Fake online point is an additional Table 1 subcondition in the case ledger, not an extra independent experiment or an imputed value in the combined-condition inventory row. Whole-service costs, training access and missing utility admit no favorable imputation or ranking.

Narrowing H2 to matched tasks leaves task-independent identity unresolved; the document proxy does not authenticate capture. H3 supports only same-record offline adaptation. The original joint workflow endpoint remains negative, including the missing-score bounds and post-outcome sensitivity views. These positive, inconclusive, zero and negative outcomes are preserved with their original endpoint

TABLE S11. ADDITIVE FAILURE/ACTIVITY DIAGNOSTICS ON ALL 1,135 TEST ROWS/226 QUESTIONS. POINT SCORES USE THE ORIGINAL QUESTION WEIGHTS; NO NEW CONFIRMATORY INTERVALS. ORIGINAL PROGRAMS ARE READ FROM SAVED PROBABILITIES, NOT REFITTED.

| Program | Loss (nats) | Accuracy (%) |
|---|---|---|
| Training prior (saved) | 2.64267 | 6.59 |
| Error/result flags only[a] | 2.47464 | 14.25 |
| Projected-empty flag only | 2.63832 | 6.79 |
| Activity/page-age only[b] | 1.81732 | 37.68 |
| Original coarse (saved) | 1.44058 | 51.29 |
| Original coarse+sequence (saved) | 1.09591 | 66.26 |

[a] Analyst-only diagnostic, not an observer-valid attacker; two binary flags, not error text. [b] Total event count and seven local-page-time summaries; no action types/order, diagnostics, text or session duration. Same four candidate architectures and validation-only selection for each of the three new representations (12 fits). Low-dimensional baselines do not rule out higher-order failure/task confounding.

TABLE S12. QUANTITATIVE ASSERTIONS EXCLUDED FOR MISSING BINDINGS.

| Source | Missing binding for quantitative use |
|---|---|
| AgentPrint | Referenced supplementary mitigation result, procedure and population not located. |
| Weiss | No matched vendor-defense before/after outcome located; the artifact appendix reproduces an attack table. |
| Gu | No provider-change result with a matching comparison population located. |
| PromptEcho | MSS/threshold padding is described; exact threshold, population, cost denominator and defended outcome are unbound. |

Sources: [30], [31], [32], [33]. No missing value is imputed; withholding these assertions does not remove the four families.

TABLE S13. SCOPED INTERVENTION DISPOSITIONS, ALL 43 CURRENT CORE FAMILIES. CATEGORIES ARE MUTUALLY EXCLUSIVE AT FAMILY LEVEL, NOT A COUNT OF SUCCESSFUL DEFENSES.

| Source-scoped disposition | Families |
|---|---|
| Evaluated mitigation component/condition | 16 |
| Fixed common guardrail (not paired installation) | 2 |
| Other intervention / different security objective | 4 |
| No evaluation identified in inspected scope | 13 |
| Diagnostic goal: confidentiality defense inapplicable | 4 |
| Unbound assertions excluded from effect comparisons | 4 |
| Total | 43 |

The four statements occur in their sources: AgentPrint supplementary analysis, Weiss vendor confirmation, Gu provider changes and PromptEcho padding figures. Quantitative result/run bindings remain unverified; all four families remain. Source-reported response, measured protection and missing measurement are distinct.

meanings.

## S7. Retained Workflow Comparison

This finite model-assisted task compares a complete flat checklist (R), complete active structured checklist (T) and SCOPE receiver traversal (V). All receive the same evidence, ten checks and output schema in fresh `gpt-6-luna`/low sessions, with randomized request order within each root. The 512 primary roots use eight constructed templates, three queries per root, 11 nodes and four configurations. They are not independent natural-paper reviews or authenticated researcher participants.

The original joint endpoint requires the decision, exact configuration counts, evidence identifiers and schema all to pass. Correctness is 40.63% (R), 45.12–45.31% (T) and 32.16% (V). One T root is unscored; ranging its whole score over $[0, 1]$ gives V-minus-T of $-13.15$ to $-12.96$pp. Under fixed request-slot potential outcomes and no cross-request interference, the independent root contrasts lie in $[-1, 1]$. The two-sided finite-assignment Hoeffding radius is

$$h = \sqrt{2\log(2/0.05)/512} = 0.12004035.$$

The union over the missing score yields the conditional 95% envelope $[-25.16, -0.95]$pp. It is not a population-of-researchers interval; provider-state interference is unverified. All 1,590 scheduled slots, including 18 secondary source-file roots, were admitted; 1,589 are scorable, with the original overload HOLD and unknown usage retained.

Preserved negative result and analysis. V has 695 decision-right/count-wrong judgments versus T's 476, with no scorable schema/evidence-ID failure. Decision-only $+1.11$–$+1.30$pp changes the endpoint and cannot rescue the joint result. Six of eight templates have negative V-minus-T effects. Post-outcome flag-free and equality-pattern reductions remain negative ($-12.43$pp and $-12.56$ to $-12.29$pp); these are descriptive sensitivity views, not replacements. The 18 secondary source-file roots give $-9.26$pp descriptively. Exact-count semantics, repeated templates and fixed graph scale limit generalization.

The companion full protocol retains assignments, root/query records, missingness arithmetic, all decompositions and input checks. These data support no advantage for the tested workflow and no claim about human productivity or universal framework inferiority.

## S8. Result-Pair and Six-Check Audit

This retrospective audit applies the paper's comparison rules to the existing 33 result records across 22 families. It adds explicit source judgments and deterministic aggregation, not new measurements or independent human coding. The machine-readable inventory retains every unordered pair ($\binom{33}{2} = 528$), the two proposed transfer directions, the coded reason and both source locators. Missing transfer evidence remains unresolved; known endpoint or mechanism conflicts are distinguished from missingness.

### S8.1. Relevance and inference definitions

Six mechanism-question groups define 102 related pairs: traffic-envelope mitigation (66), cache-service mitigation (28), behavioral humanization (3), generator fingerprint variation (3), local-memory mitigation (1), and fixed guardrail/instruction variation (1). Access control, contention noise and filter-created observations remain singleton contextual groups. The other 426 pairs are outside these specified questions, rather than silently excluded as failures. These are retrospective relevance bins, not a sample of all possible comparisons or a literature-wide denominator.

A descriptive comparison is licensed when the target and receiver-facing outcome align while each metric, population and execution mode remains explicit. It licenses discussion, not numerical ranking. A paired effect additionally requires common defended test units and a verified fitting/matching relation. Directional transfer asks whether a source operating point can be reused unchanged at the destination condition. An absent bridge is unresolved; a demonstrated wrong object or conflicting operating point blocks that particular reuse. Pairs can therefore receive different outcomes for the three inferences.

The 13 same-family pairs provide the least arbitrary comparison subset (Table S15). Seven compare aligned outcomes and six substitute different primitives or cost objects. Of the seven aligned-outcome pairs, four have known differences in collection/execution populations and three lack a verified paired-test/fitting binding. None licenses subtraction across those two records. This is separate from the positive paired adaptation contrasts already recorded *inside* UI-delay-offline and DANGOL-capture. For unchanged deployment transfer, five same-family pairs have a positively conflicting requested reuse (10 directions) and eight lack a bridge (16 directions). Across the 102 related pairs, the 204 directions comprise 10 blocked and 194 unresolved cases. No cross-record positive bridge is established by this inventory; source-local permissions remain usable.

For example, transformed Dangol captures and its live loopback stack report different operating points (approx-

TABLE S14. LOAD-BEARING SOURCE ABLATION. ENTRIES IDENTIFY ACTUAL DEPENDENCE, NOT ROBUSTNESS TO ARBITRARY MISSING STUDIES.

| Conclusion / source locator | What is supported | What is lost on source removal / NR dependence |
|---|---|---|
| Action preservation / UI Traces 6.1,Fig5 | Recorded delays preserve structure and allow adaptation | Action-delay instance and one D1/D2 positive lost; exact release-point binding not needed for protocol claim. |
| Procedural preservation / SIGLEAK 4/C | Skill inference persists under common extraction defense | Skill-defense instance lost; no paired installation-effect claim. |
| Re-emission / NetEcho IV-C–E | Downstream reserialization can restore fine granularity | Only direct downstream re-emission instance lost; no multisource replication claimed. |
| State change / AHB 5.2.3/Table1/E.3 | Fake actions can cause unintended UI transitions | Only executed mobile state-change instance lost; detector-run binding remains U. |
| Private-intent cover / WRA 8.1/Table8 | Real/decoy execution with interleaved-domain inference evaluation lowers one-decoy $E_{\mathrm{func}}$ | Private-intent decoy instance lost; AHB state-effect instance remains. Same-condition terminal utility/cost are NR-R. |
| Mode split / Dangol §9.7, OHTTP paragraph after Table 5; 9.8–9.9/Table6 | Capture frontier differs from live protocol and timing ablation | Concrete accuracy 0.17/0.30 pair and one D1/D2 positive lost. |
| Cost boundary / Early Bird V-A/B; Splice VII-C; Speculation 6.1; KVGov 5.9 | Separate component and deployment costs | Removing one withdraws its example, not other boundary instances; full service costs remain unknown. |
| Generator dependence / UI Traces, FP-Agent, traffic benchmark, LeakyLMs | Task, browser, topology and calibration tests differ | Each source's axis withdrawn when removed; no pooled portability estimate. |

TABLE S15. DESCRIPTIVE COMPARISON WITHIN DECLARED SUBSETS. PERCENTAGES HAVE THE DISPLAYED ROW DENOMINATOR, NOT ALL 528 PAIRS.

| Subset | Compatible | Incompatible | Unresolved |
|---|---|---|---|
| Same family ($n = 13$) | 7 (53.8%) | 6 (46.2%) | 0 |
| Related ($n = 102$) | 16 (15.7%) | 80 (78.4%) | 6 (5.9%) |

imately 0.17/16.2× bytes versus 0.30/13× bytes/11× latency). They can be described together as workflow-recovery evidence but not as a paired deployment improvement. SpliceDefense's CTBF signal suppression and QCP-only cost likewise answer different component questions. The counts expose the consequences of the declared comparison, not an observed rate of community misreading or a benchmark of the field schema.

## S8.2. Six checks and source corrections

Exactly the 16 families with an evaluated-mitigation disposition contribute six cells each. Table S16 summarizes these 96 judgments; the CSV retains the result IDs, source scopes, reasons and separate utility/cost objects. A partial item preserves both supported and missing elements, rather than treating a useful component measurement as no evidence.

The joint deployment item is partial for all 16 because the recorded component, replay or separately collected results do not jointly bind every required deployment object. This does not impose an agent-level experiment on studies of individual components. AHB binds detectability and objective task success but not execution cost; Dangol's live stack binds inference and protocol costs but not objective task utility; Speculation reports a padding score and token-payload ratio. In the adaptation item, partial status also reflects unbound observer access or fitted-test details even when source-local protected fitting is established.

TABLE S16. SIX CHECKS OVER 16 EVALUATED-MITIGATION FAMILIES. COUNTS APPLY TO THE SOURCE-SCOPED CLAIM; NO COMPOSITE QUALITY SCORE IS COMPUTED.

| Check | Bound | Partial | NR-V | NA |
|---|---|---|---|---|
| 1. Target/protection relation | 16 | 0 | 0 | 0 |
| 2. Receiver fields/permissions | 13 | 3 | 0 | 0 |
| 3. Placement/action dispatch | 13 | 2 | 1 | 0 |
| 4. Protected fit/test/access | 0 | 4 | 5 | 7 |
| 5. Tested workloads/shifts | 14 | 2 | 0 | 0 |
| 6. Joint inference/utility/cost | 0 | 16 | 0 | 0 |

Bound: the required item is source-bound for the scoped claim. Partial: supported elements coexist with an unbound element. Check4 includes the observer's training-data access and matching, not just existence of a fitted model. Check6 asks for an end-to-end deployment join, not a requirement that every component study perform one. No whole-check cell is solely NR-R; individual unreported subfields remain NR-R in the 96-cell inventory.

The audit corrects four historical interpretations through an additive overlay. Speculation uses a Random Forest, and Time Will Tell fits input-length-conditioned thresholds; their defended-fit/run bindings are NR-V, not inapplicable merely because the attack was described as matching or a count rule [34], [35]. SafeKV Figure 12 explicitly uses a five-second post-trigger window, resolving that field. SafeKV also reports separate serving-performance experiments: in the stated multi-turn comparison against full isolation, Qwen's extra TTFT falls from 50.41% to 11.74%, and DeepSeek's from 118% to 34.28%; at 16 requests/s its tested configurations achieve 1.36–2.66× the token throughput of Cache-Partition [36]. These are service measurements, not only detector latency. They retain their own workload/condition rather than becoming a joint leakage–cost point with Figure 12. Original frozen records remain available alongside the corrected overlay.

## S8.3. Reproduction and interpretation

The package's `decision-audit/` directory contains the rules, explicit semantic coding, all-pair inventory, 1,056 directional rows, 96 checklist cells, source corrections and read-only checker. The checker verifies unique coverage, input hashes, source/result ownership and reproducibility of the saved counts without fitting or network access. Original full texts are not redistributed; URLs, versions, locators and hashes permit source lookup. Counts are descriptive: no significance test, pooled efficacy score, causal effect, reviewer-error rate or independent human agreement is estimated. Thus the audit quantifies what the source bindings permit while keeping the underlying evidence and comparison question visible.

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